



Data structures & Algorithms

Tutorial - 01

Answer all questions.

1. Compare and contrast linear data structures Vs Nonlinear data structures using suitable examples.
2. Consider the following data array.

Array [] = {5,12,23,52,67,89}

- a. Draw a (i) flowchart diagram and write a (ii) pseudocode algorithm to print all numbers in the array.
- b. Write a Pseudocode algorithm to insert number '37' to index '3' to the above array.

Note: You may use a new array for the above insertion.

- c. Write a pseudo code to delete '89' from the above initial array.

Note: You may use a new array to store the outcome of the part (e)

3. Consider the following two sorted arrays.

2	14	41	70	88
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21	37	67	92	99
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a. Briefly describe how one can merge the above two sorted arrays into a third array.

Note: Your answer also should include a graphical representation of the merging.

b. Write a pseudocode and C code to merge the above two arrays together.